

Session title:

Computational Differential and Difference Algebra

Organizers:

[Vladimir Gerdt](#) (Dubna, Russia)

[Alexander Levin](#) (Washington D.C., USA)

[Daniel Robertz](#) (Plymouth, UK)

Objectives:

Algebraic differential and difference equations and systems of such equations arise in many areas of mathematics, natural sciences and engineering. One can say that difference equations relate to differential equations as discrete mathematics relates to continuous mathematics. Differential / difference computer algebra studies algebraic differential / difference equations in a constructive way that extends the methods and algorithms of commutative algebra and algebraic geometry. The main goal of the session is to consider the computational problems in differential / difference algebra to explore new constructive ideas and approaches oriented toward various applications.

Topics of the session include, but are not limited to:

Differential and Difference Equations and Systems
Differential and Difference Gröbner (Standard) and Involutive Bases
Differential and Difference Characteristic Sets
Triangular Decompositions of Differential and Difference Systems
Differential and Difference Elimination
Algorithmic Generation of Finite Difference Approximations to PDEs
Consistency and Stability Analysis of Finite Difference Approximations
Difference-Differential Polynomials and Systems
Software Packages for Differential and Difference Algebra
Applications of Differential and Difference Algebra in Mathematics and Natural Sciences

.....